

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. CHB-502: Inorganic Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Why is $[\text{Co}(\text{NH}_3)_6]^{3+}$ diamagnetic, whereas $[\text{CoF}_6]^{3-}$ is paramagnetic?
- (b) Differentiate between thermodynamic stability and kinetic stability of complexes.
- (c) What is the trans-effect? Give one example.
- (d) Draw the molecular structure of ferrocene.
- (e) Write the IUPAC name and structure of Ziegler-Natta catalyst.

Question 2.

- (a) Discuss the crystal field splitting of d-orbitals in octahedral and tetrahedral geometries. Derive the relation between their splitting parameters. [8]
- (b) Discuss the variation of lattice energies of the bivalent transition metal halides. Explain why the curve shows two maxima (humps) rather than a straight line. [6]

Question 3.

- (a) What is the Orgel energy level diagram? Describe the importance of this diagram for distinguishing octahedral and tetrahedral complexes of d^1 and d^9 systems. [8]
- (b) Define ligand field stabilization energy (LFSE). Calculate the LFSE for d^4 , d^6 , and d^7 octahedral complexes in both high-spin and low-spin states. [6]

Question 4.

- (a) State the mechanisms of substitution reactions in square-planar complexes. How do the charge on the metal center and nature of leaving group affect the rate? [8]
- (b) Discuss the homogeneous hydrogenation of alkenes using Wilkinson's catalyst. Write the catalytic cycle. [6]

Question 5.

- (a) What is haptacity? Show that the haptacity of a ligand varies from one organometallic compound to another using cyclopentadienyl complexes. [8]
- (b) Discuss the Dewar-Chat-Duncanson model of bonding in metal-alkene complexes, citing Zeise's salt as an example. [6]